

# Advanced Microeconomics

Dr. Juliane Hennecke

Nils Sulzmann

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## Problem Set 7: Mathematical solutions

### 1 Application

1) A person with  $U(w) = w^2$ , where  $w$  is the person's income, is facing the following lottery: there are four tickets sold, of which only one wins the prize of \$100.

a) Calculate the expected value ( $E(w)$ ) of the lottery.

**Solution:**

$$E(w) = \$25$$

b) Give the definition of and calculate the certainty equivalent ( $C$ ) of this lottery.

**Solution:**

$$C = \$50$$

2) A's utility function is  $U = \sqrt{w}$ , where  $w$  is her wealth. She owns a bakery that will be worth 70 or 0 dollars next year with equal probability.

a) Suppose her firm is the only asset she has. What is the lowest price  $P$  at which she will agree to sell her bakery?

**Solution:**

$$P = 17.5$$

b) Redo part (a) assuming that she has \$100 safely stored under her mattress.

**Solution:**

$$P = \$32.71$$

3) An individual plays soccer in a team in Brazil. If he does not get injured by the end of the next season, he will get a professional contract in Europe, worth 10,000 EUR. If he gets injured, he gets a contract as a firefighter in São Paulo, worth only 100 euros. The probability of getting injured is 20%. With 80% probability he will get through the season unscathed.

- a) What is his expected utility if his utility function is  $u(x) = \ln(x)$ ?

**Solution:** The expected utility of the lottery is:

$$E_{(U[x])} = 8.289$$

- b) What is the certainty equivalent for this individual to the lottery above? What is the risk premium?

**Solution:**

$$C = 3.979,85$$

$$R = 4.040,15$$